



CSC-361 – Machine Learning | Complex Computing Problem

Semester: Spring 2025

Max Marks: 4

Track: Track A – Research-Oriented Project

Mapped CLO: CLO-1

Instructor: Dr. Shafiq Ur Rehman Khan

Track A – Research-Oriented Project
Milestone 1 – Research Proposal and Literature Study

Course: CSC-361 Machine Learning	Semester: Spring 2025
Track: Track A – Research-Oriented Project	Due Date: 01 April 2025
Instructor: Dr. Shafiq Ur Rehman Khan	Max Marks: 4 / 20
Mapped CLO: CLO-1	

Mapped Course Learning Outcome (CLO)

- CLO-1: Explain the fundamental concepts of machine learning including supervised learning, unsupervised learning, and deep learning models.

Milestone Objective

Students will define a clear ML research problem, survey existing literature, identify a research gap, and propose a methodology with dataset justification. This milestone builds foundational understanding of machine learning concepts including supervised, unsupervised, and deep learning paradigms.

Required Deliverables

- Written Report (PDF, 3–5 pages)
- GitHub Repository link (if applicable)

Submission Requirements

- Clear and well-scoped problem statement
- Real-world motivation and significance of the problem
- Literature review covering minimum 10 recent research papers
- Identification of research gap from literature
- Explanation of relevant ML paradigm (supervised / unsupervised / deep learning)
- Dataset source with justification (size, relevance, availability)

Evaluation Rubric — Total Marks: 4 / 20



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Group Name / No.:

Submission Date:

Criterion	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Needs Work (25%)	Marks	CLO
Problem Statement & Motivation	Crystal-clear, well-scoped problem with compelling real-world motivation	Clear problem with adequate motivation	Problem stated but motivation is weak or generic	Vague or missing problem statement	0.75	CLO-1
Literature Review (≥10 papers)	10+ recent papers reviewed with critical synthesis and gap identification	10 papers reviewed; gap partially identified	Fewer than 10 papers; or gap not clearly identified	Minimal review; no gap identified	1.25	CLO-1
ML Paradigm Explanation	Correctly identifies and explains the appropriate ML paradigm (supervised/unsupervised/deep learning) with strong justification	Correct paradigm identified; explanation is adequate	Paradigm mentioned but not explained	No ML paradigm identified	1.0	CLO-1
Dataset Justification	Dataset fully described: source, size, relevance, and suitability confirmed	Dataset identified with partial justification	Dataset mentioned but no justification	No dataset identified	0.5	CLO-1
Report Quality & Structure	Well-organized, error-free, properly formatted report with references	Mostly organized with minor errors	Some structure present but inconsistent quality	Poorly written and disorganized	0.5	CLO-1

Marks Distribution Summary

Criterion	Max Marks	CLO
Problem Statement & Motivation	0.75	CLO-1
Literature Review (≥10 papers)	1.25	CLO-1
ML Paradigm Explanation	1.0	CLO-1
Dataset Justification	0.5	CLO-1
Report Quality & Structure	0.5	CLO-1
Total	4.00	



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Important Notes: All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.